MOYO AJAYI

Senior Data Scientist
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SUMMARY

A born AND grown Alaskan, Moyo is a passionate data scientist who integrates analytical and computational methods to share meaningful insight. Employers will be hiring a data scientist who:

- leverages almost 10 years of experience collecting, cleansing, analyzing, and reporting to provide actionable data-driven insights to a variety of stakeholders
- highly skilled in processing large (10s-100s GB) data sets with python (`pandas`, `spark`)
- can adeptly apply inferential statistics and machine learning using scikit-learn, MLOps, and other relevant packages to produce high-caliber work
- has a deep technical proficiency in language modeling and generative AI, enabling sophisticated analysis and optimization of unstructured data
- understands how to connect novel data science techniques to specific business questions

RELEVANT EXPERIENCE

Bridgestone — Aug 2022 - Present
Senior Data Scientist — Denver, CO

· Classifying Customer Reviews for Product Development
  · Spearheaded an 8+ model NLP pipeline, expediting customer feedback integration to influence development for $5M per product and improve efficiency as well as scale by reducing the previous manual process
  · Directed the technical aspects of the project by collaborating with 4+ data engineering consultants and synthesized our MLOps tasks into the final product

· Corporate Intelligence Retrieval Augmented Generation (RAG) Application
  · Created an end-to-end RAG model and evaluation framework to facilitate strategic planning and for corporate strategists by minimizing search time and promoting accurate and concise responses
  · The RAG model involved the collection of exclusive documents that required several iterations of exploring chunking methods, retrieval algorithms, and human and AI-assisted evaluation to arrive at a successful deployment

· Bridgestone GPT
  · Partnered with three colleagues to introduce a secure chat service for 350+ teammates as an alternative to the company-banned commercial chat services

ERM, Inc. — May 2021 - Jul 2022
Data Scientist — Denver, CO

· Semi-Supervised Learning to Optimize Practices at Large Peruvian Copper Mine
  · Eliminated manual tasks and created a predictive model framework that improved mine performance for the client
  · Employed non-linear dimension reduction (UMAP) and unsupervised ML methods (e.g., HDBSCAN) to effectively compress high-dimensional geochemical data into meaningful clusters

· Global Climate Forecasting Reporting Tool
  · Developed multifaceted tool to query, quantify, and automatically report current and projected data for a financial institution and its agribusiness assets
  · Utilized advanced modules to programmatically organize data queried from global climate models into concise and dynamic reports

Data Science Institute (Vanderbilt) — Jul 2020 - Dec 2020
Data Science Associate — Denver, CO

· Leveraged ML to Predict Teacher Churn for the State of Tennessee
  · Consulted with state education agency and produced supervised classification machine learning model using tidymodels (R) to evaluate and predict annual turnover for 65k+ teachers

PhD Candidate, Vanderbilt University — Jul 2017 - Aug 2023
Linking Greenhouse Gases and Volcanic Emissions with Data-Driven Strategies

- Orchestrated and implemented the scientific and logistic sampling design of more than 100+ samples of greenhouse gas measurements across two N. American volcanoes
- Ran inferential analyses to gain an understanding of the relationships between different locations within and between volcanoes
- Examined geospatial relationships between measurement sites
- Employed advanced statistical analysis to generate high-impact insight
- Published a paper the Journal of Volcanic and Geothermal Research on subset of analysis

MS Candidate, Vanderbilt University — Oct 2015 - Aug 2016
Robust Statistical Analysis of Fugitive Methane Emissions at Hydraulically Fractured Sites — Oliver Springs, TN

- Fashioned a mobile laboratory with state-of-the-art gas analyzer and accompanying equipment
- Used the mobile laboratory to identify fugitive CH_4 and $\delta^{13}CH_4$ escaping at the soil-atmosphere interface
- Implemented many variants of two-sample hypothesis (A/B) tests to separate the true amounts of normal background gases from fugitive leaks caused by hydraulic fracturing procedures

EDUCATION

PhD in Environmental Engineering — 2016 - 2024
Vanderbilt University, Nashville, TN

MS in Earth & Environmental Sciences — 2014 - 2016
Vanderbilt University, Nashville, TN

Bachelor's in Environmental Biology (concentration in Earth Science) — 2010 - 2014
Columbia University, New York, NY

TECHNICAL SKILLS & EXPERIENCE

Programming	Python: PyTorch, HuggingFace Suite, Scikit-Learn, Git, SQL
Data Management	AWS Stack (EC2, S3), Databricks (Apache Spark), Streamlit
Algorithms	NLP and Generative AI: Text and audio generation, embeddings (GPT-4/4o, Mistral Large, BGE)
	Supervised Classification & Regression (i.e., LGBM), Unsupervised Clustering: UMAP, HDBSCAN